

Introduction to Probability Theory II

1 Independence Continued

Note that our previously discussed characteristic of Independence was incomplete.

Definition: If two events A and B are **independent** then $P(A \cap B) = P(A)P(B)$.

Unfortunately, the converse is not true, so we need a better definition of **independent** events. To provide this we need to understand **conditional probability**.

2 Conditional Probability

2.1 Conditional Probability Based on Data

In basketball, we may consider six outcomes:

- missing a free throw (s_1),
- making a free throw (s_2),
- missing a two-pointer (s_3),
- making a two-pointer (s_4),
- missing a three-pointer (s_5), and
- making a three-pointer (s_6).

Given the sample space $\Omega = \{s_1, s_2, s_3, s_4, s_5, s_6\}$, let

- $A_1 = \{s_1, s_2\}$,
- $A_2 = \{s_3, s_4\}$,
- $A_3 = \{s_5, s_6\}$, and
- $S = \{s_2, s_4, s_6\}$.

In words, A_1 is a free throw attempt (whether it was a failure or a success), A_2 is a two-pointer attempt, and A_3 is a three-pointer attempt, S is a shot made (a free throw, two-pointer, or three-pointer), and S^c is a shot missed.

[Table 1](#) is the contingency table by the shot type and shot result of Michael Jordan in the 1985 NBA Season. By dividing each cell of [Table 1](#) by the total number of attempts 2,423, [Table 2](#) represents the contingency table in terms of probabilities.

Table 1: Contingency table by count

	Free Throw	2-pointer	3-pointer	
Event	(A_1)	(A_2)	(A_3)	Total
Success (S)	630	837	9	1,476
Failure (S^c)	116	788	43	947
Total	746	1,625	52	2,423

Table 2: Contingency table by probability

	Free Throw	2-pointer	3-pointer	
Event	(A_1)	(A_2)	(A_3)	Total
Success (S)	$\frac{630}{2423}$	$\frac{837}{2423}$	$\frac{9}{2423}$	$\frac{1476}{2423}$
Failure (S^c)	$\frac{116}{2423}$	$\frac{788}{2423}$	$\frac{43}{2423}$	$\frac{947}{2423}$
Total	$\frac{746}{2423}$	$\frac{1625}{2423}$	$\frac{52}{2423}$	1

There are two kinds of probability in [Table 2](#).

- **Joint probability:** $P(A_1 \cap S) = \frac{630}{2423} = 0.260$ means 26.0% of his shots were free throws made, $P(A_3 \cap S^c) = \frac{43}{2423} = 0.018$ means 1.8% of his shots were three-pointers missed.
- **Marginal probability:** $P(A_1) = \frac{746}{2423} = 0.308$ means 30.8% of his shots were free throws (whether it made or missed), $P(S) = \frac{1476}{2423} = 0.609$ means 60.9% of his shots were made (free throws, two-pointers, or three-pointers), and $P(S^c) = 1 - 0.609 = 0.391$ means 39.1% were missed.

Based on a joint probability and marginal probability, we can calculate a conditional probability. For example,

$$P(S | A_1) = \frac{P(S \cap A_1)}{P(A_1)} = \frac{630/2423}{746/2423} = \frac{630}{746} = 0.845.$$

In this case, we say that the conditional probability of S (success) given A_1 (free throw) is 0.845.

Provide an interpretation of the value:

Example: Calculate the following conditional probability.

$$P(A_1 | S) =$$

Interpret the value in the context of the problem.

Definition: For arbitrary events A and B , the **conditional probability** of A given B is defined as

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

provided that $P(B) > 0$ (i.e., B is a possible condition), and the conditional probability of B given A is defined as

$$P(B | A) = \frac{P(B \cap A)}{P(A)}$$

provided that $P(A) > 0$.

In practice, conditional probabilities can be calculated when count data are organized by a contingency table like in [Table 1](#) or [Table 2](#). In general, $P(A | B) \neq P(B | A)$.

2.2 Conditional Probability Based on Probability Model

Let h and t denote head and tail, respectively, for a coin toss. Suppose a coin is tossed three times. The sample space is given by $\Omega = \{hhh, hht, hth, htt, thh, tht, tth, ttt\}$. Assume all eight outcomes are equally likely so that the probability of each outcome is $\frac{1}{8}$.

Recall: Suppose we need two or more heads to win, and this event can be written as $W = \{hhh, hht, hth, thh\}$. Before tossing a coin, the probability of W is

$$\begin{aligned} P(W) &= P(\{hhh, hht, hth, thh\}) \\ &= P(hhh) + P(hht) + P(hth) + P(thh) \\ &= 0.125 + 0.125 + 0.125 + 0.125 \\ &= 0.5. \end{aligned}$$

Suppose h occurred for the first toss. Given this outcome, what is the probability of winning? The condition (h for the first toss) can be written as $H_1 = \{hhh, hht, hth, htt\}$.

2.3 Independence and Conditional Probability

Now we can define **Independent Events** using conditional probability.

Definition: We say two events A and B are **independent** when $P(A | B) = P(A)$. In other words, A and B are independent when the condition of B does not change the probability of A .

Definition: We say two events A and B are **dependent** when $P(A | B) \neq P(A)$. In other words, A and B are dependent when the condition of B changes the probability of A .

Example: Suppose t occurred for the second toss. Given $T_2 = \{hth, htt\}$, the conditional probability of W is

Are W and T_2 dependent or independent?

The new conditional definition of **Independence** is the best way to determine if two events are independent, but it is good to remember that if two events are independent then $P(A \cap B) = P(A)P(B)$.

Let's write a short proof:

3 Partition of Sample Space

The United States Senate consists of 100 Senators, and let Ω be the set of 100 Senators who can vote. On February 5, 2020, the Senators voted whether the President was guilty of abusing power (denoted by G) or not (G^c). [Table 3](#) is the vote result by the three political parties: Democratic (D), Republican (R), and Independent (I).

Table 3: Vote result summary (count)

	Demographic	Republican	Independent
Event	(D)	(R)	(I)
Guilty (G)	45	1	2
Not Guilty (G^c)	0	52	0

Since a Senator must belong to one and only one party, $D \cap R = \emptyset$, $D \cap I = \emptyset$, and $R \cap I = \emptyset$. In this case, we say D , R , and I are **pairwise disjoint**.

In addition, $\Omega = D \cup R \cup I$. In other words, the union of the three political parties is the entire Senate.

When events (groups) are pairwise disjoint and their union is the sample space Ω , we say that the events are a **partition** of Ω . Therefore, D , R , and I are a partition of Ω .

Example: Identify another partition of* Ω from the table above.

Definition: For a given sample space Ω , we say events $A_1, A_2, \dots, A_m$ are a partition of Ω when

(1) the events are pairwise disjoint and

(2) their union is Ω .

Example: Show that an event and its complement are always a partition of their sample space.

4 Law of Total Probability

As D , R , and I are a partition of Ω , the probability of G is the sum of the three joint probabilities as follows:

$$\begin{aligned}P(G) &= P(G \cap D) + P(G \cap R) + P(G \cap I) \\&= 0.45 + 0.01 + 0.02 \\&= 0.48.\end{aligned}$$

Theorem: Suppose $A_1, A_2, \dots, A_m$ are a partition of the sample space Ω . For any event B (a subset of Ω),

$$P(B) = P(B \cap A_1) + P(B \cap A_2) + \dots + P(B \cap A_m).$$

It is known as the **Law of Total Probability** (or simply **LTP**).

5 Multiplication Rule

Recall that the conditional probability of an A given B is defined as

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

provided that $P(B) > 0$. Therefore, the joint probability of A and B can be written as

$$P(A \cap B) = P(A | B) \cdot P(B)$$

This rule is known as the **Multiplication Rule** (or simply **MR**).

Note: Another expression of the MR is $P(A \cap B) = P(B | A) \cdot P(A)$.

6 Weighted Average of Conditional Probabilities

In the previous example, 48 out of 100 Senators voted guilty, and it can be expressed as $P(G) = 0.48$. Since D , R , and I are a partition of the entire sample space, we can express the marginal probability of G as follows:

$$P(G) = P(G \cap D) + P(G \cap R) + P(G \cap I)$$

The first equality is by the LTP, and the second equality is by the MR. In other words, the marginal probability $P(G)$ is the weighted average of $P(G | D)$, $P(G | R)$, and $P(G | I)$ where the conditional probabilities are weighted $P(D)$, $P(R)$, and $P(I)$, respectively.

Despite the fact that all Democratic and Independent Senators voted for guilty (i.e., $P(G | D) = 1$ and $P(G | I) = 1$), their parties were not big enough to be the majority (i.e., $P(D)$ and $P(I)$ were too small).

Theorem: Suppose $A_1, A_2, \dots, A_m$ are a partition of the sample space Ω . For any event B (a subset of Ω), its marginal probability is given by

$$P(B) = P(B | A_1) \cdot P(A_1) + P(B | A_2) \cdot P(A_2) + \dots + P(B | A_m) \cdot P(A_m).$$

In words, the marginal probability of an event is the weighted average of conditional probabilities weighted by the probabilities of the conditions, as long as the conditions are a partition of the entire sample space.

Note: For any event A , the event A and its complement A^c are a partition of Ω . Therefore,

$$P(B) = P(B | A) \cdot P(A) + P(B | A^c) \cdot P(A^c).$$

7 Wrap-Up

In your groups work on the following problems. You can check your answers below. All calculations (numeric answers) must be rounded to four decimals (e.g., 0.6667 instead of $2/3$, 0.5 or 0.5000 instead of $1/2$) unless specified. Be ready to show your work on the board.

7.0.1 Problem 1

Let A and B be two events. Fill in the blanks.

- a. $P(A \cap B)$ is called the [j... p...] of A and B .
- b. $P(A)$ is called the [m... p...] of A .
- c. $P(B | A)$ is called the [c... p...] of the event B given the possible condition A .

7.0.2 Problem 2

Let $\Omega = \{1,2,3,4,5,6\}$ for tossing a six-sided die. Assume all six outcomes are equally likely. Let $A = \{1\}$. Let O be the event of observing an odd face. Let E be the event of observing an even face. Calculate the following.

- a. $P(1)$
- b. $P(O)$
- c. $P(A | O)$
- d. $P(A | E)$
- e. $P(O | A)$

7.0.3 Problem 3

Let A and B be two events. True or false?

- a. $P(A | B) = P(B | A)$
- b. $P(A) \geq P(A | B)$
- c. $P(B) \geq P(A | B)$
- d. $P(A | B) \geq P(A \cap B)$
- e. $P(A) \geq P(A \cap B)$
- f. $P(A) + P(A^c) = 1$

7.0.4 Problem 4

Let Ω be a sample space, and A , B , and C be events. Fill in the blanks.

- a. If $A \cap B = \emptyset$, $A \cap C = \emptyset$, and $B \cap C = \emptyset$, we say that A , B , and C are a [p... d...].
- b. If A , B , and C are pairwise disjoint, and their union is Ω , we say that A , B , and C are a [p...] of Ω .
- c. A and A^c are a [p...] of Ω .

7.0.5 Problem 5

Let Ω be a sample space, and A , B , and C are a partition of Ω . Calculate the following.

- a. $P(A) + P(B) + P(C)$
- b. $P(A) + P(A^c)$

7.0.6 Problem 6

Suppose 47% of the population are female nonsmokers, 43% are male nonsmokers, 2% are female smokers, and 8% are male smokers. Suppose we randomly select a person from the population. Calculate:

- a. the probability of selecting a smoker
- b. the probability of selecting a male

7.0.7 Problem 7

It rains on Monday with a probability of 0.25. If it rains on Monday, it rains on Tuesday with a probability of 0.5. If it does not rain on Monday, it rains on Tuesday with a probability of 0.9. Calculate:

- a. the probability of raining on both Monday and Tuesday
- b. the probability of raining on Tuesday

7.0.8 Problem 8

Fill in the blank.

- a. The [L... of T... P...] is applied in Problems 6a and 6b.
- b. The [M... R...] is applied in Problems 7a.
- c. The [w... a...] of two conditional probabilities is used in Problems 7b.

7.0.9 Problem 9

It is estimated that 40%, 11%, 45%, and 4% of Americans have blood type A, B, O, and AB, respectively. It is estimated that 7%, 8%, 9%, and 7% of blood type A, B, O, and AB, respectively, develop a certain disease. Calculate the prevalence of the disease in the population. In other words, if we randomly select an American, what is the probability that the subject has the disease?

8 Assignment

Check your answer by inputting them below their respective question. All calculations (numeric answers) must be rounded to four decimals (e.g., 0.6667 instead of $2/3$, 0.5 or 0.5000 instead of $1/2$) unless specified.

Though you are checking the numeric answers here, you should make sure your written work is complete and your mathematical notation is correct as both will be graded on the homework quiz.

8.0.1 Problem 1

Suppose a student must earn a high school grade point average (HS GPA) of 2.0 or above to apply to a university. The population of interest is all students who take Calculus I in the university. Let G_i be the HS GPA group of $[i, i + 1)$ for $i = 0, 1, 2, 3, 4$ where $[a, b)$ is an interval between a and b including a and excluding b . Let A, B, C, D , and F be the groups of students who earned letter grades of A, B, C, D, and F in Calculus I, respectively. Answer the following question given the data in [Table 4](#).

Table 4: Contingency table of HS GPA and Calculus I grade (count)

Event	G_0	G_1	G_2	G_3	G_4	Total
A	0	0	12	146	16	174
B	0	0	43	216	9	268
C	0	0	44	167	3	214
D	0	0	33	48	1	82
F	0	0	32	56	1	89
Total	0	0	164	633	30	827

- Calculate the probability of earning an A in Calculus I.
- If a student's HS GPA is between 2 and 2.99, calculate the probability of earning an A in Calculus I.
- If a student's HS GPA is 4 or above, calculate the probability of earning an A in Calculus I.
- Suppose a passing grade is A, B, or C. Calculate the probability of passing Calculus I.
- If a student's HS GPA is between 2 and 2.99, calculate the probability of passing Calculus I.
- If a student's HS GPA is 4 or above, calculate the probability of passing passing Calculus I.
- If a student's HS GPA is 4 or above, calculate the probability of not passing passing Calculus I.
- If a student passed Calculus I, calculate the probability that the student's HS GPA is 3 or above.

8.0.2 Problem 2

Autism is a developmental disorder. Autistic children have difficulties with social interactions, verbal and nonverbal communications, and repetitive behaviors. According to the Centers for Disease Control and Prevention (CDC) in 2021, about one in 27 boys and one in 116 girls are identified with autism in the United States. It is estimated that the ratio of boys is about 1.05 in the United States. Let A be autism and B be a boy.

- Calculate $\frac{P(A|B)}{P(A|B^c)}$ (called the odds).
- Interpret the ratio of the two conditional probabilities in the context.
- Table 5 is a contingency table of 10,000 hypothetical children. Fill in each cell. Round to whole numbers.

Table 5: Contingency table of autism and sex (count)

Event	A	A^c	Total
B	i	ii	iii
B^c	iv	v	vi
Total	vii	viii	10,000

- If a child is autistic, calculate the probability that the child is a boy.
- Are A and B independent? Justify.

8.0.3 Problem 3

Let Ω be a sample space. Let A be an event, and suppose C_1 , C_2 , and C_3 are a partition of Ω . Assume $P(A) = 0.5$. Calculate the following.

- $P(\Omega)$
- $P(A \cap C_1) + P(A \cap C_2) + P(A \cap C_3)$
- $P(C_1) + P(C_2) + P(C_3)$
- $P(A | C_1) \cdot P(C_1) + P(A | C_2) \cdot P(C_2) + P(A | C_3) \cdot P(C_3)$

8.0.4 Problem 4

Suppose 2% of CSUMB students are Mathematics Majors. Among Mathematics Majors, 30% are seniors. What percent of CSUMB students are mathematics seniors? (Do not include the percent sign when you input the numeric answer.)

8.0.5 Problem 5

Among vaccinated people, 25% have the flu. Among non-vaccinated people, 75% have the flu. What percent of people should be vaccinated to maintain a prevalence of 30% in the population? (Do not include the percent sign when you input the numeric answer.)

8.0.6 Problem 6

If a woman smokes during pregnancy, the probability of a premature birth is 0.1. If a woman does not smoke during pregnancy, the probability of a premature birth is 0.05. If 7 in 100 pregnant women smoke on average, what is the probability of a premature birth in the population? (A probability is a number between zero and one.)

8.0.7 Problem 7

Titanic was a British passenger liner. It sank in the North Atlantic Ocean in the morning of April 15, 1912. The cause was collision with an iceberg, and many passengers lost lives.

titanic.csv is a dataset which includes information about 1,316 passengers. There are five columns:

- **id**: passenger identification number
- **class**: passenger class (first, second, or third)
- **age**: passenger's age group (adult or child)
- **sex**: passenger's sex (female or male)
- **survived**: passenger's survival status (no or yes)

To use R, first load the packages needed for data analysis.

```
library(ggformula) #graphics
library(mosaic)    #summary statistics
library(tidyverse) #data management
```

Then, we want to read in the data.

```
titanic <- read_csv( "3-intro-probability-ii/data/titanic.csv",
                    show_col_types = FALSE) ### read data, hide messages
about columns
head(titanic, 5) ### print the top 5 lines of the data

# A tibble: 5 × 5
  id class age  sex  survived
<dbl> <chr> <chr> <chr> <chr>
1     1 first adult male   yes
2     2 first adult male   yes
3     3 first adult male   yes
4     4 first adult male   yes
5     5 first adult male   yes
```

We can use two functions: `count()` and `props()` to evaluate our specified categories, $\sim x$.

```
counts( ~survived, data = titanic ) #counts

  response n_no n_yes
1 survived 817  499

props( ~survived, data = titanic) # proportions

  response  prop_no prop_yes
1 survived 0.6208207 0.3791793
```

We can use the formula $\sim x \mid z$ to calculate conditional counts and proportions of x conditioned on z .

```
### count by groups
counts(~ survived | sex, data = titanic )

  response  sex n_no n_yes
1 survived female 123  324
2 survived  male  694  175

### proportions survived by groups
props(~ survived | sex, data = titanic)

  response  sex  prop_no prop_yes
1 survived female 0.2751678 0.7248322
2 survived  male  0.7986191 0.2013809
```

The last R output can be interpreted as follows: about 27.5% of female passengers did not survive, about 72.5% of female passengers survived, about 79.9% of male passengers did not survive, and about 20.1% of male passengers survived.

Use the Law of Total Probability and the Multiplication Rule to calculate the probability of survival in three different ways. Though the code helps make the calculations efficient, write out the mathematical notation each value and calculation represents (see online for code and check).

- a. Estimate the probability of passengers survived. Answer based on the partition by passenger class (first, second, and third).
- b. Estimate the probability of passengers survived. Answer based on the partition by age group (adult and child).
- c. Estimate the probability of passengers survived. Answer based on the partition by sex (female and male).
- d. Explain why the three estimates in parts a, b, and c are the same.